NALANETRA FLOODGRID

AI-DRIVEN HYPER-LOCAL SEVERE WEATHER & URBAN FLOOD EARLY WARNING SYSTEM

KAYA'26
ORGANIZED BY AZMTH
HACKATHON • AI & AUTONOMOUS
SOFTWARE HACKATHON · PS-05

Team: Dropwise

PROBLEM STATEMENT 05: **AI-Driven Hyper-Local Early Warning System for Severe Weather Nowcasting (Cloudbursts & Flash Floods)**

Coupling Dual-Pol Doppler Radar Nowcasting (2–6h Lead Time), High-Res DEM Hydrodynamics & Proof-Gated Municipal Closure

[01] SEVERE WEATHER NOWCASTING

- **2–6 Hours Actionable Lead Time** before cloudburst/flash-flood impact
- **Dual-Pol Doppler Radar QPE** + Integrated Water Vapor (IWV) ingestion
- **Spatiotemporal Deep Learning** recognizes pre-event atmospheric signatures
- **Multi-task Neural Network** outputs simultaneous storm & flash-flood risk maps

[02] HYDRAULIC DRAINAGE COUPLING

- **High-Res DEM Topography** & Watershed Delineation mapping
- **Graph-based Drainage Network:** Manhole nodes & stormwater pipe edges
- **Kinematic-Wave Runoff Simulation** (Manning's Equation hydrodynamics)
- **Predictive Street Inundation** & Vulnerable Underpass Depth Triage

[03] GROUND-TRUTH & ACCOUNTABLE ACTION

- **1-Tap Bilingual Citizen Reporting** (GPS, Depth & Photo Verification)
- **P-Score Algorithm:** P = 0.30S + 0.20R + 0.15W + 0.15D + 0.10E + 0.10A
- **150m Spatial Clustering** & Dynamic Rerouting via OSRM Flood Nav API
- **Proof-Gated Municipal Closure:** Mandatory Before/After Tamper-Evident Photos

EVENT: KAYA'26 (Organized by AZMTH) | **TRACK:** Software Hackathon (AI & Autonomous) | **TEAM:** Dropwise | **PS-05:** Hyper-Local Severe Weather Nowcasting

NALANETRA FLOODGRID

URBAN FLOOD INTELLIGENCE & ACCOUNTABLE MUNICIPAL RESPONSE ECOSYSTEM

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PROBLEM OVERVIEW & GROUND REALITY

- **Economic Impact:** ₹15,000+ Cr annual damage across Indian urban centres.
- **Critical Delays:** 12-18 hours severe gridlock & submerged highway underpasses.
- **Blind Forecasts:** Macro weather forecasts miss hyper-local street waterlogging.
- **Ghost Closures:** 40-60% citizen drainage complaints closed prematurely.
- **Emergency Hazard:** Commuters & ambulances trapped with zero flood routing.

PROPOSED SOLUTION (NALANETRA ECOSYSTEM)

- **Predictive Layer (Doppler + DEM):** Target: 3-hr hyper-local flood-risk nowcast.
- **Ground Truth Validation:** 1-Tap citizen reporting with GPS & depth tags.
- **Dynamic Conflict Resolution:** Visual evidence upgrades priority (Moderate -> Critical).
- **Proof-Gated Closure Loop:** Field crews submit before/after photo proof.
- **Graph Drainage Hydraulics:** Manholes (nodes) & stormwater pipes (edges) modeled.
- **Flood-Aware Navigation API:** Waterlogged road status API for flood-aware routing.

HIERARCHICAL SYSTEM ARCHITECTURE

TIER 1: ADMINISTRATION & COMMAND ➔ MCG OFFICER COMMAND CENTRE

- Interactive GIS Heatmap & Hotspot Feeds
- Explainable Priority Queue with Officer Verification & 150m Clustering
- Crew Assignment Workflow, Response Dispatch & SLA Analytics

▼

TIER 2: CORE INTELLIGENCE ENGINE ➔ NALANETRA AI & HYDRAULIC CORE

- Doppler Radar (IMD) + DEM Inundation Simulation
- Graph-based Drainage Hydraulic Model (Nodes & Edges)
- Priority Formula: P = 0.30S + 0.20R + 0.15W + 0.15D + 0.10E + 0.10A
- Flood-Safe Routing: OSRM + OSM with Dynamic Flooded Edge Penalization

▼

TIER 3: END-USER & FIELD RESPONSE ➔ CITIZEN & FIELD CREW APP

- Citizen App: 1-Tap Hindi/English Reporting, Live 5-Stage Stepper
- Field Crew App: GPS Navigation, Turn-by-Turn Flood-Safe Routing
- Proof Gate: Evidence-Linked Before/After Photo Closure Workflow

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TECHNICAL APPROACH

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COUPLING FORECAST INTELLIGENCE WITH GROUND TRUTH & PROOF-GATED CLOSURE

METHODOLOGY (8-STAGE PIPELINE)

1. Hydrological Analysis
Gurugram hotspots & MCG drainage SOPs

2. DEM & Network Modeling
Topographic elevation & manhole graph nodes

3. Doppler Radar Ingestion
IMD nowcast data & live weather feeds

4. Priority Engine Algorithm
Formula P-Score & dynamic severity upgrade

5. Multi-Portal System Dev
Citizen, MCG Officer & Field Crew App

6. Safe Navigation API
OSRM + OSM dynamically rerouting around submerged road segments

7. Pilot Ward Deployment
Single-ward deployment in Ward 14 Gurugram

8. City-Wide Scaling
Multi-ward rollout & repeatable municipal model

INCIDENT LIFECYCLE FLOW CHART & PRIORITY FORMULA

PHASE A: DETECTION & TRIAGE

1. Rain Nowcast / Citizen Report
2. AI Priority & 150m Clustering
3. Officer Crew Dispatch

PHASE B: ACTION & VERIFICATION

4. Flood-Safe Crew Navigation
5. Choke Cleared & Photo Proof
6. Proof-Gated Verified Close

SCIENTIFIC PRIORITY FORMULA (MATHEMATICAL MODEL):

$$P = 0.30(S) + 0.20(R) + 0.15(W) + 0.15(D) + 0.10(E) + 0.10(A)$$

S: Severity Depth (Ankle to Submerged) • R: Doppler Radar Nowcast
W: Ward Criticality Index • D: Drainage Hydraulic Pipe Capacity
E: Emergency / Hospital Route (1.5x Multiplier) • A: SLA Aging & Accumulation

P-Score provides explainable prioritization; weights will be calibrated using pilot incident and response data.

TECHNOLOGY ARCHITECTURE — FULL-STACK OVERVIEW

LAYER	TECHNOLOGY	ROLE IN NALANETRA	KEY CAPABILITY
Frontend (Maps & UI)	**Flutter (Dart) + Material 3** flutter_map + Mapbox / Maptiler GL	**Interactive Flood Dashboard** Vector map rendering with smooth zoom & rotation.	**Zero-Lag Rendering** High-FPS flood heatmaps without "tile loading" errors.
AI & Intelligence	**FastAPI (Python) + OpenCV** YOLOv8 / MobileNet (On-Device)	**Real-time Severity Triage** Validates user photos locally before upload.	**Junk Filtering** Rejects fake/blurry reports instantly (saves server cost).
Data & Routing	**PostgreSQL + PostGIS** OSRM + Dynamic Weights	**Spatial Graph DB & Safe Routing** Stores road networks & flooded polygons.	**Dynamic Rerouting** OSRM graph updates live to mark flooded roads as "Blocked".
Security & Services	**SHA-256 + Secure Storage** Firebase FCM (Notifications Only)	**Evidence Integrity & Alerts** Encrypts proofs locally; Push alerts via FCM.	**Tamper-Proof Audit** Raw evidence secure even on rooted devices.

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FEASIBILITY AND VIABILITY

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MVP VALIDATION PATH • WARD 14 PILOT ➔ GROUND TRUTH ➔ PROOF-GATED CLOSURE ➔ SCALE

1. TECHNICAL
Doppler radar, DEM rasters, graph hydraulics & GPS APIs are proven technologies.

2. ECONOMICAL
Zero sensor capex; mobile & open APIs. Pilot estimate: <₹1.2L/mo vs ₹35L+ hardware.

3. SOCIAL
Bilingual UI (Hindi/English) with 1-tap workflows empowers all citizens.

4. LEGAL & COMPLIANCE
Aligned with NDMA 2010 urban flood guidelines & MoES data standards.

5. OPERATIONAL
Human-in-the-loop: automated AI triage with full officer dispatch control.

6. SECURITY & AUDIT
Role-based access, SHA-256 evidence hashes & tamper-evident audit trails.

COMPETITIVE EDGE MATRIX (Why NalaNetra Beats Existing Systems)

Key Capability	NalaNetra (Ours)	Municipal 1800	Google FloodHub
Drainage-Rain Coupling	✓ Dynamic Coupling	✗ Manual	✗ Basin Only
Hyper-Local Street Truth	✓ 1-Tap GPS Truth	✗ Voice Calls Only	✗ None
Proof-Gated Photo Closure	✓ Mandatory Proof	✗ High Ghost Risk	✗ None
Flood-Aware Safe Routing	✓ Flood-Aware Routing	✗ No Routing	▲ Partial
Specialized Hardware Cost	✓ ZERO (Mobile)	✗ High Sensor Cost	✗ High Sensor Cost

POTENTIAL CHALLENGES & MITIGATION STRATEGIES

- **Monsoon Connectivity Blackout:** Offline storage (FlutterSecureStorage) queues photos; auto-syncs online.
- **Fake / Spam Citizen Reports:** AI photo-relevance screening with mandatory officer verification before dispatch.
- **Duplicate Report Overhead:** Auto-clusters reports within 150m and 6 hours into one incident ID.
- **Weather API Rate Limits:** Local caching of Doppler rasters with resilient heuristic fallback.

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IMPACT AND BENEFITS

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CREATING ACCOUNTABLE, PROOF-BACKED AND RESILIENT URBAN DRAINAGE GOVERNANCE

TARGET: 80% FASTER
Rapid Response
Pilot goal: reduce response from 180 min baseline toward <35 min.

PROOF-LINKED
Proof-Linked Closures
Before/after evidence and location checks reduce unverifiable closures.

TARGET: -70%
Optimized Logistics
150m/time-window grouping designed to reduce redundant crew trips.

TARGET: 3 HOURS
Lead Time Goal
Lead time to be calibrated using weather, terrain and citizen inputs.

Enhanced Public Safety
Protects residents, commuters and emergency corridors in high-risk urban locations.

Hospital Route Protection
Designed to maintain clear transit corridors for ambulances and critical healthcare access.

Municipal Accountability
Evidence-linked and auditable workflow from complaint to verified sign-off.

Citizen Trust & Transparency
Live 5-stage status stepper eliminates public communication blackouts.

Resilient Urban Planning
Historical inundation analytics inform long-term municipal drain upgrades under AMRUT 2.0.

3-PHASE IMPLEMENTATION & NATIONWIDE SCALING ROADMAP

PHASE 1: PILOT WARD
Months 1–3 • Ward 14 & NH-48, Gurugram

- Coupled Doppler + DEM hyper-local nowcast engine
- Up to 25 response crews for coordinated field operations
- Bilingual Hindi/English citizen reporting app
- Real-time underpass flood depth triage

PHASE 2: CITY-WIDE EXPANSION
Months 4–8 • City-wide expansion after pilot validation

- Progressive ward-by-ward rollout based on pilot outcomes
- Dynamic OSRM flooded edge rerouting
- Live API link with Traffic Police feeds
- Automated contractor SLA enforcement

PHASE 3: NATIONAL SCALING
Months 9–18 • Replication to other cities after validation

- Inter-city expansion under AMRUT 2.0
- MoES / NCMRWF unified radar pipelines
- Standardized open APIs for NDMA framework
- Replicable smart-city resilience model

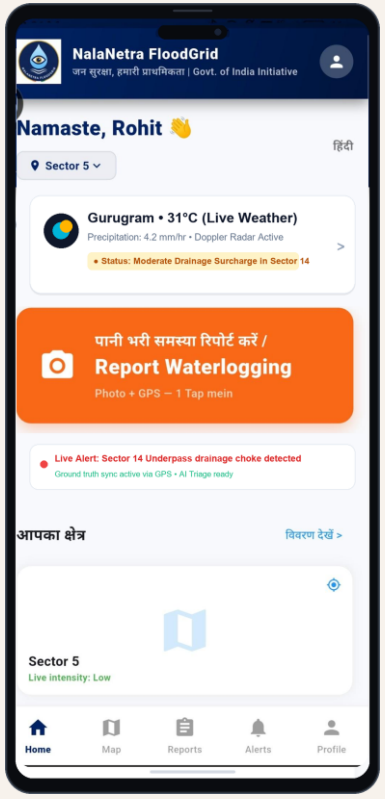
RESEARCH & REFERENCES

Team: Dropwise

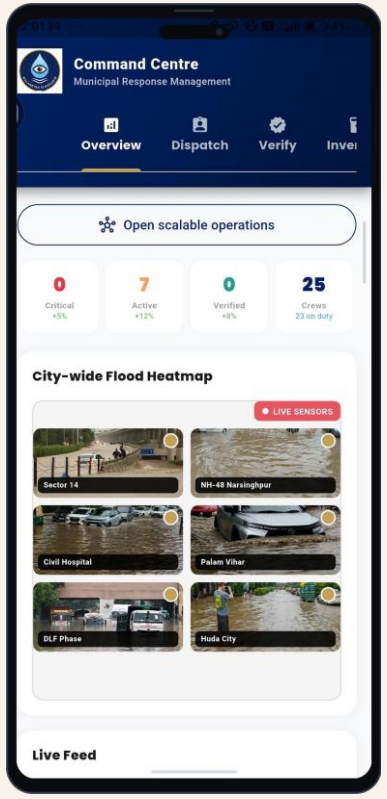
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WORKING SYSTEM IMPLEMENTATION & STATUTORY GOVERNANCE ALIGNMENT

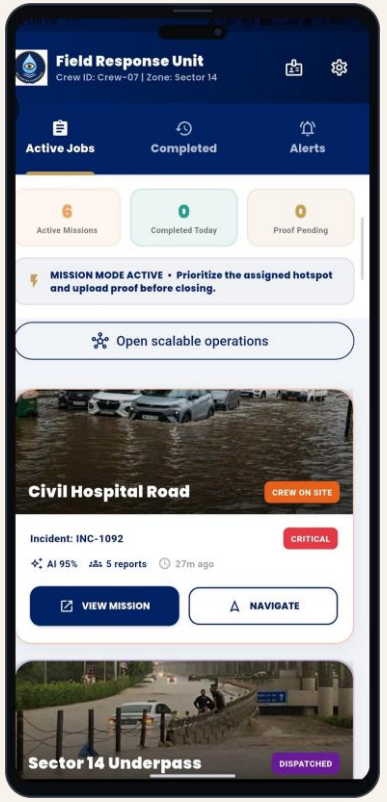
OUR WORKING PROTOTYPE



Citizen Portal (Hindi UI)
1-Tap GPS Report, Live Weather
& 5-Stage Stepper



MCG Officer Portal
City Heatmap, P-Score Priority
Queue & Crew Dispatch



Field Response Unit
Turn-by-Turn Safe Navigation
& Mandatory Photo Proof

SCIENTIFIC RESEARCH & METHODOLOGICAL FOUNDATIONS

1. Dual-Pol Doppler Radar Quantitative Precipitation Estimation (QPE)

• **Method:** Optical-flow vector tracking & Z-R calibration from Doppler radar. [NalaNetra Role: Sub-km rain volume feeds for nowcast engine.] [→ View Radar Research]

2. 1-D Kinematic-Wave Drainage Routing (Manning's Equation)

• **Method:** Hydrodynamic stormwater pipe conveyance & manhole surcharge (EPA-SWMM). [NalaNetra Role: Couples surface rainfall with drainage capacity.] [→ View EPA-SWMM Model]

3. Monocular Computer Vision Severity Triage & Evidence Integrity

• **Method:** Automated waterlogging depth classification & SHA-256 upload integrity. [NalaNetra Role: Triages citizen reports & flags inconsistent evidence.] [→ View CV Research Paper]

STATUTORY GOVERNANCE & NATIONAL REFERENCES

• **National Disaster Management Authority (NDMA):** 2010 Urban Flood Guidelines — Stormwater drainage, early warning & municipal SOPs. [→ Open NDMA Guidelines]

• **India Meteorological Department (IMD):** Urban Nowcasting SOP — Standard operating doctrine for extreme heavy rainfall events. [→ Open IMD Nowcast]

• **NCMRWF • MoES:** Unified Model (UM) — High-resolution atmospheric nowcasts coupled with drainage networks. [→ Visit NCMRWF]

• **National Institute of Disaster Management (NIDM):** Urban Flood Management — Standard training doctrine for municipal drainage resilience. [→ Open NIDM Report]